

Infrastructure as Code

Software Architecture

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- Do a quick poll.
- Who has heard the term IaC before this course?
- Who has used IaC before this course?

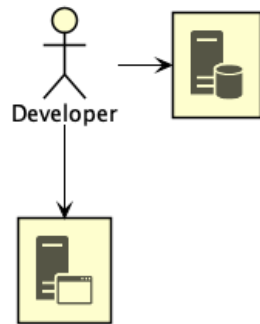
Infrastructure as Code

How did we get here?

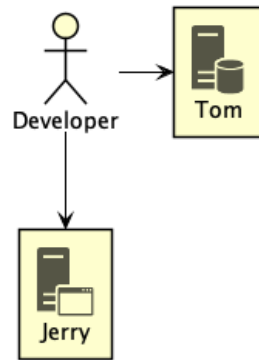
Pre-2000

The *Iron Age*

Iron Age

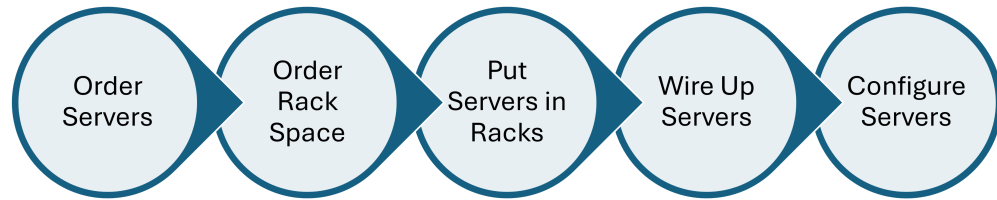


Iron Age



Developer only had a few machines — so few the machines often got fun names.

Scaling

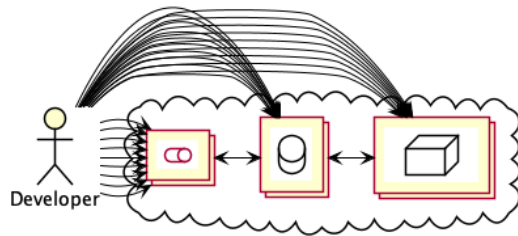


I used to have scheduling examples for software projects, where ordering servers was on the critical path, and had to happen before software development started.

Introducing...

The *Cloud Age*

The Cloud Age



- Things got complicated quickly
 - We need more hardware and it is easier to provision
- Largely thanks to virtualisation
 - No physical activity to spin up a new machine

When faced with complexity

Automate it!

- We have too much to manage to do it manually
- We're about to start enumerating automation techniques

The larger story

Server Config Config Management

The larger story

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Application Config Config Files

The larger story

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Application Config Config Files

Provisioning Infrastructure Code

The larger story

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Building Continuous Integration

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Specifications Behaviour Driven Development

Definition 0. Infrastructure Code

Code that provisions and manages *infrastructure resources*.

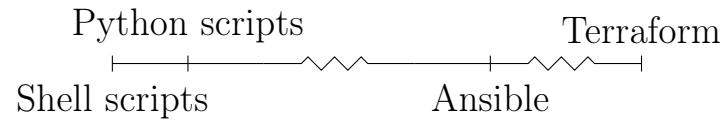
Definition 0. Infrastructure Code

Code that provisions and manages *infrastructure resources*.

Definition 0. Infrastructure Resources

Compute resources, networking resources, and storage resources.

Infrastructure Code



IC often thought of as the right-hand side, but includes all

Shell Scripts

```
1  #!/bin/bash

3  SG=$(aws ec2 create-security-group ...)

5  aws ec2 authorize-security-group-ingress --group-id "$SG"

7  INST=$(aws ec2 run-instances --security-group-ids "$SG" \
8      --instance-type t2.micro)
```

Using AWS CLI to create EC2 access like in the practical

Python

```
1  import boto3

3  def create_instance():
4      ec2_client = boto3.client("ec2", region_name="us-east-1")
5      response = ec2.create_security_group(...)
6      security_group_id = response['GroupId']

8      data = ec2.authorize_security_group_ingress(...)

10     instance = ec2_client.run_instances(
11         SecurityGroups=[security_group_id],
12         InstanceType="t2.micro",
13         ...
14     )
```

Using AWS Python library (boto3)

Terraform

```
1 resource "aws_instance" "hextris-server" {
2     instance_type = "t2.micro"
3     security_groups = [aws_security_group.hextris-server.name]
4     ...
5 }

7 resource "aws_security_group" "hextris-server" {
8     ingress {
9         from_port = 80
10        to_port = 80
11        ...
12    }
13    ...
14 }
```

Finally, Terraform

Question

Notice anything different?

- Prompting for declarative
- Might notice verbosity

The main difference

Imperative vs. Declarative

- *Imperative* – Describe the steps to take to deploy the infrastructure
- *Declarative* – Describe the desired infrastructure
- IC is heading towards a more *declarative* paradigm

Declarative IaC

- Define your *desired* infrastructure state
 - As code
- Engine interprets difference between the *desired* and *actual* state
 - Modifying infrastructure to deliver *desired* state

Infrastructure Code

- Provisions and manages *infrastructure resources*

Infrastructure Code

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- Ranges from simple shell scripts up to...?

Infrastructure Code

- Provisions and manages *infrastructure resources*
- Only one part of the movement to *automate* the complexities of development
- Ranges from simple shell scripts up to...?
- Tendency to be *declarative*

Summarising what we've already covered

Typo?

Infrastructure Code $\neq$ Infrastructure *as* Code

- Mention that this distinction is ours
- Real world unfortunately mixes the two

Definition 0. Infrastructure as Code

Following the same *good coding practices* to manage Infrastructure Code as standard code.

Warning!

Infrastructure as Code still *early* and quite *bad*

- Code reuse is low
- Importing existing resources is non-trivial
- Refactoring is painful
- State management can be tricky

Question

What are *good coding practices*?

Ask the class

Good Coding Practice #1
Everything as Code

A practice we do but barely discuss in ‘regular’ programming because it doesn’t make sense not to do it

```
1  #!/bin/bash
3  ./download-dependencies
4  ./build-resources
5  cp -r output/* artifacts/
```

```
1  #!/bin/bash
3  ./download-dependencies
4  ./build-resources
5  cp -r output/* artifacts/
```

```
$ cp: directory artifacts does not exist
```

An example of relying on external state in ‘regular’ programming

```
1 resource "aws_instance" "hextris-server" {  
2     instance_type = "t2.micro"  
3     security_groups = ["sg-6400"]  
4     ...  
5 }
```

Draw a parallel to the bash example and this, which relies on ‘sg-6400’ existing

```
1 resource "aws_instance" "hextris-server" {
2     instance_type = "t2.micro"
3     security_groups = [aws_security_group.hextris-server.name]
4     ...
5 }

7 resource "aws_security_group" "hextris-server" {
8     ingress {
9         from_port = 80
10        to_port = 80
11        ...
12    }
13    ...
14 }
```

The better approach

Everything as code avoids
Configuration drift

Configuration drift creates
Snowflakes

- *Snowflakes* magical machines that ‘just work’ and everyone is afraid to touch
- Snowflake because they’re unique and easy to break, because no one knows how it works

Benefits

1. Reproducible

Good Coding Practice #2

Version Control

Benefits

1. Restorable
2. Accountable

Good Coding Practice #3
Automation

Benefits

1. Consistent

Automatically applying or checking IC is in sync means the main branch is consistent with reality

Good Coding Practice #4

Code Reuse

Benefits

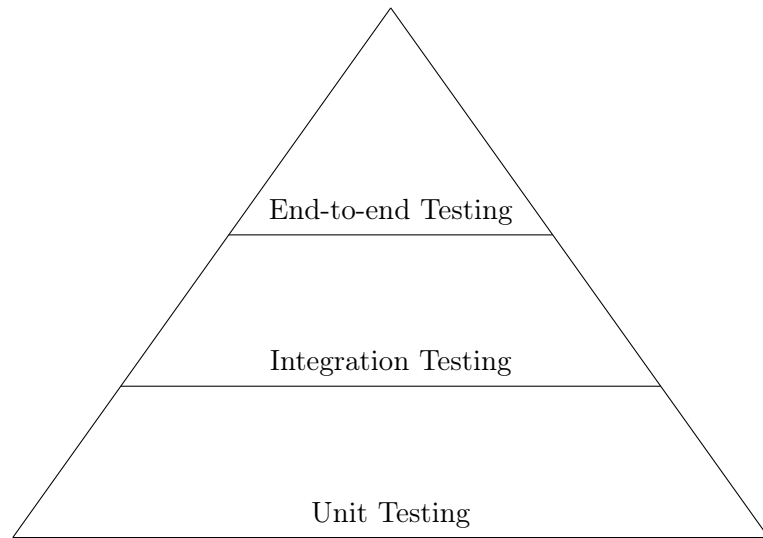
1. Better¹ code
2. Less work
3. Only one place to update (or verify)

¹generally

Good Coding Practice #5

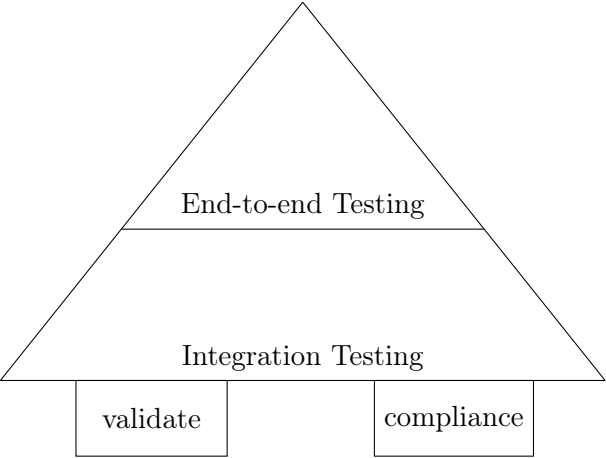
Testing

Test Pyramid



- Traditional test pyramid
- Unit testing relies on isolated testing
- But...isolated testing doesn't make *much* sense for IaC

IaC Test Pyramid



```
1 func TestTerraformAwsInstance(t *testing.T) {
2     terraformOptions := terraform.WithDefault(t, &terraform.Options{
3         TerraformDir: "../week03/",
4     })
5
6     defer terraform.Destroy(t, terraformOptions)
7     terraform.InitAndApply(t, terraformOptions)
8
9     publicIp := terraform.Output(t, terraformOptions, "public_ip")
10    url := fmt.Sprintf("http://%s:8080", publicIp)
11
12    http_helper.HttpGetWithCustomValidation(t, url, nil, 200,
13        func(code, resp) { code == 200 &&
14            strings.Contains(resp, "hextris")})
15 }
```

Validation example

```
1 Feature: Define AWS Security Groups
```

```
3 Scenario: Only selected ports should be publicly open
```

```
4     Given I have AWS Security Group defined
```

```
5     When it contains ingress
```

```
6     Then it must only have tcp protocol and port 22,443 for 0.0.0.0/0
```

Compliance testing example

Benefits

1. Trust

Week 4 Prac

Learn how to use Terraform to write IaC and deploy resources on AWS.